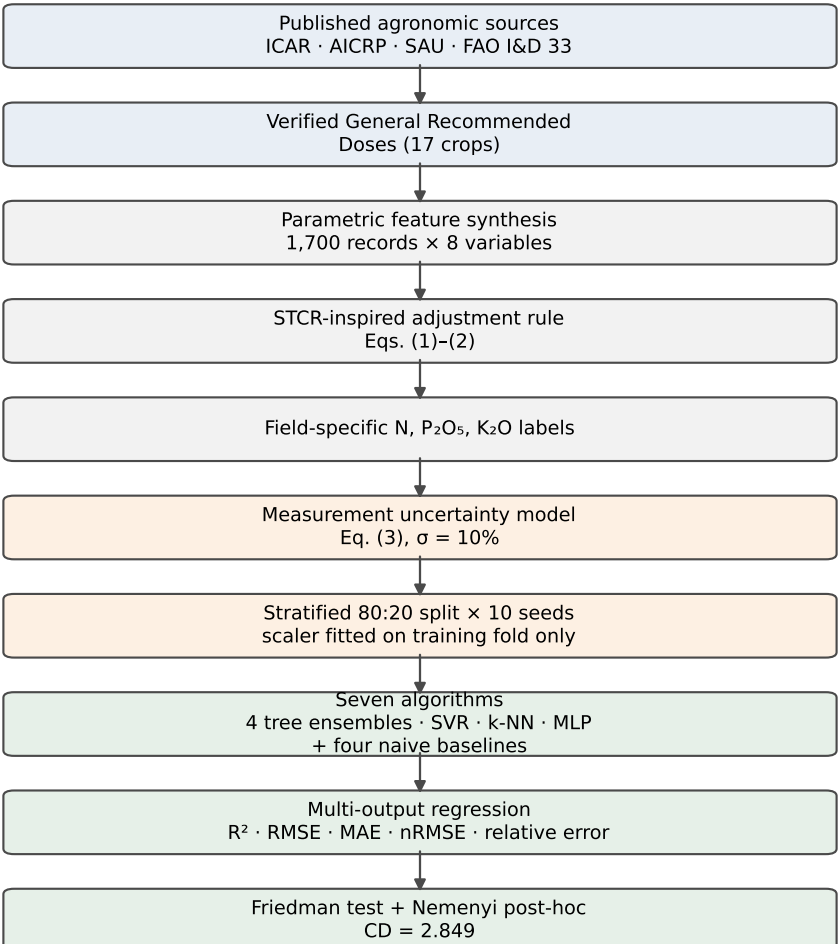
Published agronomic sources
ICAR · AICRP · SAU · FAO I&D 33
Verified General Recommended
Doses (17 crops)
Parametric feature synthesis
1,700 records × 8 variables
STCR-inspired adjustment rule
Eqs. (1)–(2)
Field-specific N, P2O5, K2O labels
Measurement uncertainty model
Eq. (3), σ = 10%
Stratified 80:20 split × 10 seeds
scaler fitted on training fold only
Seven algorithms
4 tree ensembles · SVR · k-NN · MLP
+ four naive baselines
Multi-output regression
R² · RMSE · MAE · nRMSE · relative error
Friedman test + Nemenyi post-hoc
CD = 2.849
Field validation — future work